

## ICT2223 – Internet of Things

### Week2: IoT Ecosystems and Architecture

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#### Session Objectives

By the end of this lesson, students should be able to:

1. Explain the concept of an IoT ecosystem.
2. Describe the IoT reference models and architectural layers.
3. Explain the role of:
  - Sensors
  - Actuators
  - Controllers/processors
  - Connectivity
  - Gateways
  - Edge computing
  - Storage
  - Data abstraction
  - Applications
  - Business processes
4. Identify technologies and protocols used at different IoT layers.
5. Explain how different IoT components communicate and work together.
6. Explain the relationship between **IoT, Big Data and analytics**.
7. Describe how IoT data moves from the physical world to applications.
8. Explain major IoT security requirements.
9. Identify major challenges in designing IoT ecosystems.
10. Design a basic IoT architecture for a real-world problem.

### Question

1. A digital thermometer
2. A smartwatch
3. A Wi-Fi smart bulb
4. A CCTV camera
5. A soil-moisture sensor connected to an ESP32
6. A factory machine that predicts failures

**Which of these are IoT devices? Why?**

### Answers

| Device                                 | IoT Device?                       | Why?                                                                                             |
|----------------------------------------|-----------------------------------|--------------------------------------------------------------------------------------------------|
| <b>1. Digital thermometer</b>          | <b>Usually No</b>                 | It measures temperature but does not necessarily communicate data over a network.                |
| <b>2. Smartwatch</b>                   | <b>Yes</b>                        | It collects data and communicates with smartphones/cloud services through wireless connectivity. |
| <b>3. Wi-Fi smart bulb</b>             | <b>Yes</b>                        | It connects to a network and can be monitored or controlled remotely.                            |
| <b>4. CCTV camera</b>                  | <b>Yes (if network-connected)</b> | A network camera collects video data and can transmit it to other systems or the cloud.          |
| <b>5. Soil-moisture sensor + ESP32</b> | <b>Yes</b>                        | The sensor collects environmental data and the ESP32 can transmit it through Wi-Fi/Bluetooth.    |

|                                                  |            |                                                                                                   |
|--------------------------------------------------|------------|---------------------------------------------------------------------------------------------------|
| <b>6. Factory machine that predicts failures</b> | <b>Yes</b> | It collects machine data, communicates the data, and uses analytics to predict maintenance needs. |
|--------------------------------------------------|------------|---------------------------------------------------------------------------------------------------|

### **What makes something an IoT system?**

An IoT system generally involves:

**Physical world → sensing → processing → communication → data → decision → action**

A useful modern definition:

**IoT is a system of physical objects equipped with sensing, processing, communication and/or actuation capabilities that enables them to collect, exchange, process and act on data through networks.**

### **Important distinction**

**A sensor alone is not necessarily an IoT system.**

For example: Temperature sensor → measures temperature.

But:

Temperature sensor → ESP32 → Wi-Fi → MQTT → cloud → dashboard → alert  
is an **IoT system**.

## **1. WHAT IS AN IoT ECOSYSTEM?**

### **IoT Ecosystem**

#### **Simple definition**

**An IoT ecosystem is the complete collection of devices, software, networks, platforms, data, services, users and business processes that work together to achieve an IoT objective.**

## **Think beyond “devices”**

An IoT ecosystem contains:

### **Things**

- Sensors
- Actuators
- Embedded devices
- Machines

↓

### **Connectivity**

- Wi-Fi
- Bluetooth/BLE
- Ethernet
- Zigbee
- Thread
- LoRaWAN
- Cellular

↓

### **Computing**

- Microcontrollers
- Gateways
- Edge devices

- Cloud computing



## **Data**

- Databases
- Data streams
- Data lakes
- Time-series data



## **Intelligence**

- Rules
- Analytics
- Machine Learning
- Edge AI



## **Applications**

- Mobile apps
- Web dashboards
- Control systems



## **People & Processes**

- Users
- Engineers
- Managers
- Automated workflows
- Business decisions

## **Modern perspective**

An IoT ecosystem is therefore not simply:

**Sensor + Internet**

It is:

**Physical world + Connectivity + Computing + Data + Intelligence +  
Applications + People/Processes**

## **2. IoT REFERENCE MODELS**

**A reference model is a conceptual way of organizing an IoT system. It is not a rule saying every IoT system must contain exactly seven physical layers.**

### **IoTWF 7-LAYER MODEL**

IoTWF stands for IoT World Forum

The IoTWF Reference Model has 7 layers:

1. Physical Devices & Controllers
2. Connectivity
3. Edge Computing
4. Data Accumulation
5. Data Abstraction
6. Application
7. Collaboration & Processes

### **Layer 1 — Physical Devices & Controllers**

**What is here?**

- Sensors

- Actuators
- Microcontrollers
- Embedded devices
- Machines
- Cameras
- Smart appliances

This is the layer containing the “things” of IoT and identifies data generation as its primary function.

## **Sensors**

Sensors convert physical phenomena into measurable data.

Examples:

| <b>Physical phenomenon</b> | <b>Sensor</b>        |
|----------------------------|----------------------|
| Temperature                | Temperature sensor   |
| Humidity                   | Humidity sensor      |
| Light                      | LDR/light sensor     |
| Distance                   | Ultrasonic sensor    |
| Motion                     | PIR sensor           |
| Pressure                   | Pressure sensor      |
| Gas                        | Gas sensor           |
| Soil moisture              | Soil moisture sensor |

## **Actuators**

Actuators do the opposite:

**They convert a control signal into a physical action.**

Examples:

- LED
- Motor
- Servo motor
- Relay
- Buzzer
- Valve

**Important concept**

**Sensor = Observe**

**Actuator = Act**

**This is a very useful distinction.**

## **Layer 2 — Connectivity**

The Connectivity Layer (Layer 2) is responsible for connecting IoT devices and moving data between devices, networks, edge systems, and the cloud.

In simple terms:

It answers the question: “How does the data travel?”

- Ethernet
- Wi-Fi
- Bluetooth Low Energy
- Zigbee
- Thread
- LoRaWAN
- NB-IoT
- LTE-M
- 4G/5G

A comparison among communication technologies

| Technology   | Typical strength                    |
|--------------|-------------------------------------|
| Wi-Fi        | High data rate / local connectivity |
| BLE          | Very low power / short range        |
| Zigbee       | Low power mesh                      |
| Thread       | Low-power IP mesh                   |
| LoRaWAN      | Long range / low power              |
| Cellular IoT | Wide-area connectivity              |
| Ethernet     | Reliable wired connection           |

### **Modern addition: Matter + Thread**

For smart homes, introduce **Matter**.

Matter is an IP-based smart-home application standard designed to improve interoperability across ecosystems. It can operate over Ethernet, Wi-Fi and Thread, with Bluetooth Low Energy used for commissioning.

Matter 1.5, released in 2025, expanded support to areas including cameras, closures, soil sensors and energy-management capabilities.

### **Layer 3 — Edge Computing**

This layer is processing data close to the source, reducing latency and network load through filtering, aggregation and transformation.

#### **What is Edge Computing?**

**Edge computing processes data near where it is generated instead of sending everything to a distant cloud.**

#### **Example**

Suppose a factory camera generates:

**10,000 video frames per minute.**

Instead of sending every frame to the cloud:

**Camera → Edge AI device → Detect abnormal event → Send only important information**

**Edge can perform:**

- Filtering
- Aggregation
- Compression
- Transformation
- Local decision-making
- Real-time alerts
- AI/ML inference

**Why Edge Computing?**

- **Lower latency** – Data is processed close to the IoT device, so there is less delay in getting a response.
- **Less bandwidth** – Only necessary data is sent to the cloud, reducing the amount of network traffic.
- **Faster response** – Local processing allows IoT devices to make decisions quickly without waiting for the cloud.
- **Better resilience when Internet connectivity is poor** – Devices can continue processing and operating locally even when the Internet connection is slow or unavailable.
- **Potentially better privacy** – Sensitive data can be processed locally without sending all of it to a remote cloud server.

**Edge AI = AI inference performed close to the data source.**

**Example:**

**Camera → Edge AI → “Person detected” → Cloud receives event**

**rather than:**

**Camera → Cloud → AI processing → Decision**

**Activity- CLOUD OR EDGE?**

- A.** A smart door lock must respond immediately.
- B.** A company wants to analyse five years of machine data.
- C.** A camera detects whether someone entered a restricted area.
- D.** A smart meter sends monthly electricity statistics.

Ask:

**EDGE / CLOUD / BOTH?**

| Scenario                                           | Best Choice                                                                                        | Why?                                                                         |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <b>A. Smart door lock must respond immediately</b> | <br><b>Edge</b> | Immediate local processing provides a faster response with very low latency. |

|                                                             |                                                                                                   |                                                                                                                   |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>B. Analyse five years of machine data</b>                | <br><b>Cloud</b> | The cloud provides large-scale storage and powerful computing for historical analysis.                            |
| <b>C. Camera detects someone entering a restricted area</b> | <br><b>Edge</b>  | Processing the video locally allows immediate detection and reduces the need to send video to the cloud.          |
| <b>D. Smart meter sends monthly electricity statistics</b>  | <br><b>Cloud</b> | Monthly statistics do not require immediate responses, so data can be sent to the cloud for storage and analysis. |

### Simple rule

**Need an immediate response? → Edge**

**Need large-scale storage/analysis? → Cloud**

**Need both local speed + cloud storage/analysis? → Both**

### Layer 4 — Data Accumulation

The **Data Accumulation Layer** is responsible for **collecting and storing data** received from IoT devices and networks.

#### In simple terms:

It answers the question: “**Where do we store the data so it can be used later?**”

- Converts **data in motion** into **data at rest** by storing it.
- Collects data from many IoT devices.
- Stores data for **future analysis, reporting, and decision-making**.
- Can use **databases, cloud storage, or other data-storage systems**.

#### Example:

A smart factory collects temperature readings from 1,000 sensors and stores them in a cloud database for later analysis.

IoT devices continuously generate streams.

The system must decide:

- What should be stored?
- Where should it be stored?
- For how long?
- In what format?

Examples:

- SQL databases
- NoSQL databases
- Time-series databases
- Object storage
- Data lakes
- Cloud storage

## **Layer 5 — Data Abstraction**

The **Data Abstraction Layer** organizes and processes stored IoT data so that it can be **easily accessed and understood by applications**.

**In simple terms:**

It answers the question: **“How can we make the stored data useful and accessible?”**

- Combines and organizes data from different sources.
- Converts data into a **consistent and usable format**.
- Provides data to applications through **APIs or other interfaces**.
- Hides the complexity of the underlying data storage and systems.

**Example:**

Temperature data stored in different databases is organized into a common format so a **smart factory dashboard** can easily display it.

Activities include:

- Data cleaning
- Normalization
- Aggregation
- Data integration
- APIs
- Data models
- Metadata
- Data access

## **Layer 6 — Application**

**The Application Layer provides the IoT services and applications that users interact with.**

**In simple terms:**

**It answers the question: “What can users do with the IoT data?”**

- **Uses processed data to provide useful services and functions.**
- **Presents information through dashboards, mobile apps, alerts, or control systems.**
- **Allows users or systems to monitor and control IoT devices.**

**Example:**

**A smart home application displays the home's temperature and allows the user to turn the air conditioner on or off.**

**Examples:**

- Mobile applications
- Web dashboards
- Smart-home applications

- Industrial dashboards
- Monitoring systems
- Alert systems

### **Is the dashboard itself IoT?**

It is part of the IoT ecosystem, but the complete IoT system extends from physical devices through communication, computing and data processing to the application.

### **Layer 7 — Collaboration & Processes**

**The Collaboration & Processes Layer connects IoT applications with people, business processes, and organizational decision-making.**

**In simple terms:**

**It answers the question: “How do we use IoT information to take action and improve processes?”**

- **Helps people make decisions based on IoT data.**
- **Integrates IoT information into business workflows and processes.**
- **Supports collaboration between people, systems, and departments.**

**Example:**

**A factory detects a machine problem → alerts the maintenance team → creates a maintenance task → the team fixes the machine and updates the system.**

### **Example**

IoT detects:

Machine temperature = abnormal

Application shows:

 Machine overheating

But a real organization needs:

Alert → Maintenance ticket → Technician assigned → Machine inspected →  
Repair completed → Record updated

So IoT ultimately supports **business processes and decisions**.

## THE COMPLETE IoTWF STORY

### 1. Things



### 2. Connect



### 3. Process near the source



### 4. Store



### 5. Make data usable



### 6. Build applications



### 7. Take action / support decisions

## 3. WHY DO WE NEED A REFERENCE MODEL?

1. Decompose the IoT problem into smaller parts.
2. Identify technologies at each layer and how they relate.
3. Allow different components to be provided by different vendors.
4. Define interfaces that support interoperability.

5. Define security across transitions between levels.

### **Modern interpretation**

A reference architecture helps answer:

**“Where does this technology belong, and how does it interact with the rest of the system?”**

## **4. THREE-LAYER MODEL**

The **3-Layer IoT Model** is one of the **simplest and most commonly used ways to explain how an IoT system works**.

It divides an IoT system into three main layers:

1. **Perception Layer** – Collects data from the physical world using sensors and interacts with devices.
2. **Network Layer** – Transfers data between devices, gateways, servers, and other systems.
3. **Application Layer** – Uses the collected data to provide useful services to users.

**In simple terms:**

**Perception → Collect data**

**Network → Move data**

**Application → Use data**

### **1. Perception Layer**

**Physical sensing and data capture**

Examples:

- Sensors
- RFID
- Cameras

- Actuators

## **2. Network Layer**

### **Communication and transmission**

Examples:

- Wi-Fi
- Bluetooth
- Ethernet
- Cellular
- LoRaWAN

## **3. Application Layer**

### **User-specific IoT services**

Examples:

- Smart home
- Smart agriculture
- Smart healthcare
- Smart city

**Easy memory trick**

**SEE → SEND → SERVE**

**Perception → Network → Application**

## **5. FIVE-LAYER MODEL**

**The 5-Layer IoT Model provides a more detailed view of an IoT system than the 3-layer model. It separates data processing and business activities into their own layers.**

**The five layers are:**

- 1. Perception Layer** – Collects data from the physical world using sensors and devices.
- 2. Transport Layer** – Transfers data between devices, processing systems, and networks.
- 3. Processing Layer** – Stores, processes, and analyses the collected data.
- 4. Application Layer** – Provides IoT services and applications to users.
- 5. Business Layer** – Uses IoT information to support business decisions, processes, and overall system management.

**In simple terms:**

**Perception → Collect**

**Transport → Move**

**Processing → Process**

**Application → Provide services**

**Business → Make decisions**

**Modern interpretation**

| <b>Layer</b> | <b>Main question</b>            |
|--------------|---------------------------------|
| Perception   | What is happening physically?   |
| Transport    | How does data move?             |
| Processing   | Where is data processed?        |
| Application  | What service is provided?       |
| Business     | What decision/value is created? |

## Activity 3 — ARCHITECTURE MAPPING

### Smart Agriculture System

Components:

- Soil moisture sensor
- ESP32
- Wi-Fi
- MQTT
- Edge gateway
- Database
- ML model
- Mobile app
- Irrigation pump
- Farmer
- Irrigation policy

Place each component into:

**Perception → Transport → Processing → Application → Business**

## 6. **IoT ECOSYSTEM AS A WHOLE**

**A successful IoT ecosystem must balance:**

**Technology**

Can we build it?

**Business**

Is it economically viable?

## **Users**

Does it solve a real problem?

## **Operations**

Can it be maintained?

## **Security**

Can it be trusted?

## **Sustainability**

Can it operate efficiently over its lifetime?

## **7. FUNCTIONAL BLOCKS**

The major functional blocks of an IOT eco system includes;

### **1. Sensing**

Collect information.

Examples:

- Temperature
- Moisture
- Gas
- Motion
- Light

### **2. Processing**

Embedded processors transform raw sensor data into useful information.

Examples:

- Microcontrollers
- SoCs

- ESP32
- STM32
- Raspberry Pi
- Edge computers

Tasks:

- Filtering
- Aggregation
- Encryption
- Local decisions
- AI inference

### 3. Gateway

An **IoT Gateway** acts as a **bridge between IoT devices and other networks or cloud systems**.

**In simple terms:**

**IoT Gateway = Connects, processes, and manages communication between devices and the outside network.**

#### **Main tasks**

- **Connect devices** – Connects sensors/devices using different communication technologies.
- **Protocol translation** – Converts between different protocols, such as **Zigbee → IP** or **Modbus → MQTT**.
- **Data filtering** – Removes unnecessary data before sending it to the cloud.
- **Local processing** – Can process data locally for faster responses.

- **Security** – Helps authenticate devices and protect data communication.
- **Cloud connectivity** – Sends relevant data from devices to cloud platforms.

### **Example**

Zigbee Sensor



IoT Gateway



MQTT (A lightweight way for IoT devices to send and receive messages)



Cloud

## **4. Application**

Applications provide:

- Monitoring
- Visualization
- Control
- Notifications
- Automation

Examples:

- Mobile app
- Web dashboard
- Industrial dashboard
- Smart agriculture platform

## **8. IoT COMMUNICATION**

**IoT communication** refers to the methods and technologies used by IoT devices to **exchange data with other devices, gateways, servers, and cloud platforms.**

**In simple terms:**

**IoT Communication = How IoT devices talk to each other and to the Internet.**

**Ex:**

- Wi-Fi
- Ethernet
- BLE
- Zigbee
- Thread
- LoRaWAN
- Cellular

from:

**Application/messaging protocols**

- MQTT
- CoAP
- HTTP/HTTPS

## **9. APPLICATIONS OF IoT**

**Smart Homes**

- Lighting
- Energy efficiency
- Security cameras

**Smart Cities**

- Traffic monitoring

- Waste management
- Pollution tracking

### **Smart Healthcare**

- Remote patient monitoring
- Wearable health trackers

### **Smart Agriculture**

- Precision farming
- Soil moisture sensing

### **Smart Logistics**

- Fleet tracking
- Inventory monitoring

### **Smart Energy**

- Grid optimization
- Demand response

### **Modern examples**

- Smart buildings
- Industrial IoT
- Predictive maintenance
- Connected vehicles
- Environmental monitoring
- Digital twins
- Energy-aware smart homes
- Edge-AI surveillance systems

## **Activity 4— DESIGN AN IoT SYSTEM**

### **Group A**

Smart classroom

### **Group B**

Smart greenhouse

### **Group C**

Smart parking

### **Group D**

Smart waste bin

### **Group E**

Factory predictive maintenance

You must identify:

1. What should be sensed?
2. What sensor is needed?
3. What controller?
4. What connectivity?
5. Edge or cloud?
6. What data is stored?
7. Which architecture will be used?
8. What application?
9. What action should happen?
10. Who uses the system?
11. What could go wrong?

## **10.IoT + BIG DATA**

IoT generates massive volumes of heterogeneous data and that analytics converts this data into actionable insights.

**IoT generates the data.**

**Big Data technologies manage the scale.**

**Analytics extracts patterns and insights.**

**AI/ML can make predictions or decisions.**

## **11. WHAT IS BIG DATA?**

### **Volume**

Large quantities of data.

### **Velocity**

Data arrives rapidly.

### **Variety**

Different forms of data.

Examples include sensor readings, logs, images and video.

### **Modern characteristics**

**Veracity** – How trustworthy is the data?

**Value** – Can we obtain useful value from it?

**5 Vs: Volume, Velocity, Variety, Veracity, Value**

## **12.IoT DATA IS NOT ALWAYS “BIG DATA”**

A single temperature sensor sends one reading every hour. Is that automatically Big Data?

**No.**

IoT can produce Big Data when the combination of:

- Number of devices
- Data frequency
- Data types
- Data duration
- Processing requirements

creates significant scale and complexity.

## **13.TYPES OF ANALYTICS**

### **1. Real-Time Analytics**

Immediate analysis.

Example:

Detect machine overheating.

### **2. Offline Analytics**

Historical analysis.

Example:

Analyse machine failures over five years.

### **3. Memory-Level / In-Memory Analytics**

Data is processed in memory for faster operations.

### **4. Business Intelligence Analytics**

Supports decision-making and reporting.

## **5. Massive Analytics**

Complex processing over very large datasets.

These are all in the original material.

### **Activity -WHAT TYPE OF ANALYTICS?**

1. “The machine temperature reached 95°C.”
2. “The machine overheated because cooling efficiency decreased.”
3. “The machine is likely to fail within 24 hours.”
4. “Reduce machine load and schedule maintenance.”

### **Answers**

1. “The machine temperature reached 95°C.”  
**Descriptive Analytics – Tells us what happened.**
2. “The machine overheated because cooling efficiency decreased.”  
**Diagnostic Analytics – Explains why it happened.**
3. “The machine is likely to fail within 24 hours.”  
**Predictive Analytics – Predicts what is likely to happen.**
4. “Reduce machine load and schedule maintenance.”  
**Prescriptive Analytics – Recommends what action should be taken.**

## **14.IoT BIG DATA PROCESSING PIPELINE**

PHYSICAL WORLD



Sensors / Devices



Connectivity



Edge Processing



Message Broker / Ingestion



Storage



Stream + Batch Processing



Analytics / AI



Dashboard / API



Decision



Actuator



PHYSICAL WORLD

**CLOSED-LOOP IoT**

**Sense → Understand → Decide → Act**

Example:

**Soil moisture sensor**



Detects dry soil



Analytics determines irrigation is required



Controller decides



Pump turns ON



Soil becomes sufficiently moist



Pump turns OFF

## **15.IoT ARCHITECTURE FOR BIG DATA**

**IoT DEVICES**

Sensors / Actuators / Cameras



**CONNECTIVITY**

Wi-Fi / Ethernet / BLE / LoRaWAN / Cellular



EDGE / GATEWAY

Filtering / Protocol Translation / Edge AI



DATA INGESTION

MQTT / HTTP / Streaming



DATA PLATFORM

Time-Series DB / NoSQL / Object Storage



ANALYTICS

Stream Analytics / ML / AI



APPLICATION

Dashboard / Mobile App / API



BUSINESS PROCESS

Alert / Decision / Automation



ACTUATOR

Physical Action

## **16. SECURITY IN IoT**

- Device security
- Network security
- Application security

- Data security

**Security is not something added at the end.**

It should exist from:

**Design → Development → Deployment → Operation → Update → Retirement**

NIST's April 2026 revision specifically emphasizes cybersecurity activities across the IoT product lifecycle, including maintenance, support and end-of-life considerations.

**Activity — “Find the Security Weakness”**

Scenario:

An ESP32-based smart door system:

- uses a default password
- communicates without encryption
- never updates firmware
- exposes an API to the Internet
- stores user data indefinitely

**Identify five security weaknesses and propose one improvement for each.**

**Answer**

| <b>Security Weakness</b>   | <b>Why it is a problem</b>                                  | <b>Improvement</b>                                     |
|----------------------------|-------------------------------------------------------------|--------------------------------------------------------|
| <b>1. Default password</b> | Attackers may know or easily guess the default credentials. | Require a <b>strong, unique password</b> during setup. |
| <b>2. No encryption</b>    | Data transmitted between the ESP32 and other                | Use <b>encrypted communication</b> such as             |

|                                         |                                                                  |                                                                                   |
|-----------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|                                         | systems could be intercepted.                                    | <b>TLS/HTTPS or MQTT over TLS.</b>                                                |
| <b>3. No firmware updates</b>           | Known vulnerabilities may remain unpatched.                      | Provide <b>regular, signed firmware updates.</b>                                  |
| <b>4. API exposed to the Internet</b>   | Direct Internet exposure increases the attack surface.           | Restrict access using <b>authentication, authorization, and network controls.</b> |
| <b>5. User data stored indefinitely</b> | Unnecessary data retention increases privacy and security risks. | Apply <b>data minimization and automatic deletion/retention policies.</b>         |

## 17.IoT CHALLENGES

### 1. Scalability

Managing millions of devices.

### 2. Interoperability

Integrating different devices and platforms.

### 3. Energy Efficiency

Power management for battery-powered devices.

### 4. Data Management

Storing and processing large-scale streaming data.

### 5. Security & Privacy

Protecting sensitive user and operational data.

### 6. Device Lifecycle Management

How do we:

- register devices?

- update firmware?
- replace devices?
- revoke compromised devices?
- retire old devices?

## **7. Data Quality**

Bad sensor data can lead to bad decisions.

**Garbage in → garbage out.**

## **8. AI Reliability**

If AI is used at the edge or cloud:

Can we trust its predictions?

## **9. Vendor Lock-in**

What happens if the IoT platform provider changes its pricing or service?

## **10. Sustainability**

Millions of connected devices create:

- battery waste
- electronic waste
- energy consumption
- hardware replacement issues

# **18.MODERN IoT CONCEPTS**

## **AIoT**

**AI + IoT**

IoT generates data.

AI extracts intelligence.

Together:

**Sense → Learn → Predict → Act**

### **Edge AI**

AI processing happens close to the device.

Example:

Camera → Edge AI → Detect object → Immediate action

### **Digital Twins**

A **digital twin** is a digital representation of a physical asset or system that can be updated using data from the physical system.

Example:

Physical factory machine ↔ Digital representation

IoT provides the data that keeps the digital representation updated.

### **IoT + Cloud**

Cloud provides:

- scalable storage
- computing
- analytics
- centralized management

### **IoT + Edge**

Edge provides:

- low latency
- local processing
- reduced bandwidth

- resilience

**Modern architecture**

**Device + Edge + Cloud**

**not:**

**Device + Cloud only.**

## **19.FINAL INTEGRATED CASE STUDY**

### **Smart Greenhouse**

A university wants to automatically control irrigation in a greenhouse.

Requirements:

- Measure soil moisture.
- Measure temperature.
- Measure humidity.
- Monitor the greenhouse remotely.
- Automatically operate the water pump.
- Store historical data.
- Alert the farmer when abnormal conditions occur.
- Predict irrigation requirements in the future.

**You must design:**

#### **Layer 1**

What sensors and actuators?

#### **Layer 2**

What connectivity?

#### **Layer 3**

What edge processing?

#### **Layer 4**

What storage?

#### **Layer 5**

How will data be abstracted?

#### **Layer 6**

What application?

#### **Layer 7**

What business/process decision?

#### **Security**

How will you secure the system?

#### **Big Data**

What happens if 10,000 greenhouses generate data continuously?

#### **AI**

Where could ML be introduced?

## **20. Questions**

### **Q1**

Is a temperature sensor by itself an IoT system? Explain.

### **Q2**

What is the difference between a sensor and an actuator?

**Q3**

Why do IoT systems need connectivity?

**Q4**

What problem does an IoT gateway solve?

**Q5**

Why is edge computing useful?

**Q6**

What is the difference between data accumulation and data abstraction?

**Q7**

Compare the 3-layer and 5-layer IoT models.

**Q8**

Why does IoT generate Big Data?

**Q9**

Explain the difference between real-time and offline analytics.

**Q10**

Where would you use AI in an IoT system?

**Q11**

Give two examples of IoT security threats.

**Q12**

Why is interoperability important?

**Q13**

Design a simple IoT architecture for a smart classroom.

**Q14**

Where would you place MQTT in an IoT architecture?

**Q15**

Explain:

**Sense → Connect → Process → Store → Analyse → Decide → Act**

## Summary

| Topic                      | Purpose                                        |
|----------------------------|------------------------------------------------|
| What is an IoT ecosystem?  | Core concept                                   |
| Components of an ecosystem | Build vocabulary                               |
| IoT reference models       | Introduce architecture                         |
| IoTWF 7-layer model        | Detailed architecture                          |
| Three-layer model          | Simplified view                                |
| Five-layer model           | More detailed view                             |
| Functional blocks          | Sensors → processors → gateways → applications |
| Connectivity               | Modern protocols                               |
| MQTT & communication       | Practical communication model                  |
| Edge computing             | Modern architecture                            |
| Edge AI                    | Current technology                             |
| IoT applications           | Real-world relevance                           |
| IoT + Big Data             | Connect IoT to data                            |
| Big Data 5Vs               | Understand scale                               |
| Types of analytics         | Understand intelligence                        |
| IoT data pipeline          | End-to-end flow                                |
| IoT + Cloud                | Scalable computing                             |
| IoT + Edge + Cloud         | Modern architecture                            |
| AIoT                       | Modern direction                               |
| Digital Twins              | Emerging concept                               |
| Security                   | Secure IoT                                     |
| IoT lifecycle security     | Modern practice                                |
| Challenges                 | Critical thinking                              |

